An Efficient Parameter Estimation and Imaging Approach for Ground Maneuvering Targets by Mixed Symmetric Function in SAR Imagery

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***Abstract*— Ground moving target focusing performance and processing efficiency are two important metrics in the synthetic aperture radar (SAR) system. Owing to the complex nonstationary phases caused by target motions, ground maneuvering targets are commonly smeared and distorted in an SAR image. To realize the ground maneuvering target imaging, exhaustive high-order parameter searching and optimization processing are usually needed to achieve the optimal processing performance. However, the large computational resource consumption may limit their real-time processing capabilities. To deal with this issue, an efficient SAR imaging approach for ground maneuvering targets is proposed by adopting the mixed symmetric function (MSF) transform. After performing the multichannel SAR clutter rejection and target initial detection by applying the constant false alarm rate (CFAR) technique, the linear envelop shift of detected ground target is eliminated by using the well-known Keystone transform (KT). Then, an MSF is constructed to rectify the range curvature and relieve the Doppler broadening after carrying out the nonuniform fast Fourier transform (NUFFT) corresponding to the second-order time variable. After removing the second-order Doppler spreading influence, the quadratic chirp parameter is estimated via the NUFFT after the symmetrical correlation processing. Finally, after accomplishing the motion compensation, the ground maneuvering target can be well focused and relocated in the SAR image. Compared with the traditional approaches, the proposed approach not only realizes the high-precision Doppler parameter estimation, but is also computationally efficient without grid searching procedure. Numerical simulation and real-measured SAR experiment results are depicted to verify the correctness and effectiveness of the proposed approach.**

***Index Terms*— Ground maneuvering target imaging, high efficiency, high-order Doppler parameter estimation, mixed symmetric function (MSF), synthetic aperture radar (SAR).**

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I. Introduction

DUE to the long virtual aperture formed by the radar platform high-speed movement, synthetic aperture radar (SAR) can effectively realize the high-precision electromagnetic imaging of ground observation scene [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], which has been extensively utilized in various fields, including forest covering mapping, natural disaster risk assessment, and maritime environment observation. Ground moving target indication (GMTI) technique, as an important expansion for SAR applications, can provide an important solution to achieve the target detection, positioning, and recognition in a finely focused SAR image, which has been a research hotspot in the research community [11], [12], [13], [14], [15].

Due to the down-looking work mode in an SAR-GMTI system, when ground target echo returns are captured by the radar receiver, ground clutter scatterers will also be irradiated by the radar mainlobe beam. Different from the conventional ground- or sea-based static radars, owning to the rapid motion of the SAR platform, the clutter components contain a broad mixture of radial line-of-sight (RLOS) velocities within the mainlobe antenna beam, causing the target signal submerged in the clutter background [16], [17], [18], [19], [20], leading to the rapid decrease of target detection probability.

Due to the easy implementation in practical engineering and low system cost, the early research works mainly concentrate on the target detection in a single-channel SAR configuration. Moreira and Keydel [21] and Fienup [22] realized the ground target detection by exploiting the time–frequency signal difference between the ground static scatterers and the uncooperative moving targets. Lv et al. [23], [24] and Wang [25] accomplished the target detection by exploiting the positive and negative center spectrum difference property. After obtaining the phase components of the target signal after range cell migration (RCM) compensation, Ryu et al. [26] accomplished the Doppler parameter estimation by using the

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least-square fitting technique. However, due to the mainlobe clutter interference, the above-mentioned single-channel-based GMTI methods may be more suitable for the fast-moving target detection in the clutter "clean area."

To improve the target detection capability in the clutter regions, multiple equivalent receiving arrays installed along the azimuth direction are necessary, so as to increase the SAR spatial-domain degree of freedom. The corresponding multichannel signal processing methods in the clutter background can be categorized into three kinds: the post-Doppler domain clutter rejection methods [27], [28], [29], [30], [31], [32], the SAR image-domain clutter suppression methods [33], [34], [35], [36], [37], and velocity SAR (VSAR) [4] and the Chinese remainder theorem (CRT), such as [6]. After performing the clutter spatial filtering, the target output signal-to-clutter and noise ratio (SCNR) will be significantly enhanced due to the target spatially coherent synthesis and clutter removal. However, because of the phase mismatch of target with unknown motion characteristics, the troublesome RCM and Doppler spreading [38], [39], [40] will appear, especially for ground maneuvering targets. Therefore, to improve the target detection performance to the utmost extent, these deleterious influences should be sufficiently dealt with.

As for the RCM compensation, based on the correlation property between adjacent receiving pulses, some inverse SAR (ISAR)-based envelop shift compensation methods can be applied to align the curved target trajectory in an SAR system, typical methods including the envelop automatic correction method [41], [42], adaptive profile peak estimation method [43], and entropy minimization method [44], [45]. In [46], [47], [48], [49], and [50], the Hough transform combined with a phase filtering function using the priori SAR parameters is proposed to realize the target RCM elimination. Due to the noncoherent signal processing, these methods may be more suitable for the envelope shift alignment of ground strong targets. In the low SCNR situation, the target trajectory may be hardly observed after range compression. At this time, Keystone transform (KT) [51] can be used to blindly accomplish the range walk compensation independence on target SCNR, which is much computationally efficient by using the scaling transform principle [52], [53]. Then the residual second-order RCM error can be approximately removed by exploiting the priori SAR knowledge [54], [55]. To further improve the second-order RCM alignment accuracy, motivated by the KT, the second-order KT is proposed based on the second-order time-scaled transform [56], [57], [58], [59], which can completely remove the range curvature for multiple ground moving targets without the requirement of target kinetic knowledge. However, these KT or modified KT-based methods may not remove, simultaneously, linear and second-order RCM errors in a high-resolution SAR-GMTI system.

After the RCM removal, the Doppler spread should also be accurately compensated so as to accomplish the cross-range imaging. As for the ground uniformly moving targets, the second-order polynomial phase signal (PPS) can be sufficiently employed to characterize the moving target phase property, see, for example, the chirp parameter estimation in [60], [61], [62], [63], and [64]. But, as for a maneuvering target during relatively long observation time, the cubic phase signal (CPS) model will be more accurate to realize the target finely imaging, and the corresponding parameter estimation algorithms can be coarsely divided into two kinds: noncorrelation algorithms [65], [66], [67] and correlation algorithms [68], [69], [70], [71], [72], [73]. The noncorrelation methods can usually obtain the relatively superior target focusing quality via the multidimensional grid search, but giving rise to the heavy computational burden, especially in a high-resolution radar system with lots of azimuth sampling pulses. Therefore, the phase-order reduced correlation methods become necessary.

Considering that the Doppler phase estimation performance is sensitive to the RCM compensation accuracy, some algorithms directly exploiting the target 2-D phase structure have been proposed to accomplish the RCM and azimuth broadening compensation together. Zhu et al. [74] deduced the 2-D coupling relationship between range frequency and Doppler frequencies in an SAR-GMTI system based on stationary phase principle (SPP), and designed an efficient filtering function independent of target kinetic parameters to realize the target imaging. To further improve the target focusing quality, the 2-D frequency searching method is introduced to remove the smearing influence caused by target along-track motions [75]. However, these SPP-based methods are only applicable to the situation of a large time–bandwidth product (TBP) along azimuth direction. In the last decade, the Radon-based coherent processing algorithms in [76], [77], and [78] were proposed to accomplish the target focusing along the complex trajectories of moving targets, exhibiting the relatively good focusing performance under the condition of low SCNR, but at the sacrifice of increasing the computational resource consuming.

Without grid searching of high-order Doppler parameters, an efficient target focusing method is proposed by exploiting the symmetrical characteristics of slow time and range frequency. In the proposed method, KT is first employed to remove the linear RCM, where the residual range curvature is unavoidably present. Then, based on the symmetrical phase structure, the mixed symmetric function (MSF) is designed to completely remove the adverse smearing effect caused by the second-order RCM, where the azimuth chirp parameter can be efficiently estimated by using the nonuniform fast Fourier transform (NUFFT) [79], [80]. In the meantime, after eliminating the symmetric smearing effect caused by the chirp parameter, the nonsymmetric Doppler broadening caused by the quadratic chirp parameter can also be effectively compensated by employing the NUFFT. Finally, the signal energy corresponding to a ground maneuvering target can be fully integrated, beneficial to the subsequent target relocation. In contrast to the traditional target focusing methods in [9], [14], [51], [52], [54], [55], [56], [57], [58], [59], [81], [83], and [84], the proposed approach can effectively deal with the sophisticated range-azimuth coupling issues caused by the high-order phase errors with relatively low computational resource consumption. Experimental results from the

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simulation data and real-measured SAR data are employed to verify the effectiveness and applicability of the proposed approach.

The other parts of this article are given as follows. In Section II, the high-order signal model with respect to a ground maneuvering target is built in a multichannel SAR system. In Section III, the detailed parameter estimation principle of the proposed approach is introduced, where the performance analyses including the computational complexity and multi-target focusing results are given. In Section IV, simulation and real-measured experimental results are provided to verify the proposed approach. Finally, some conclusions and future prospects are given in Section V.

II. Signal Model and Problem Description

Fig. 1 illustrates the 3-D observation configuration in a multichannel SAR system with a height of H, where the Cartesian coordinate of O-XYZ is built, with X-axis expressing the range dimension and Y-axis corresponding to the azimuth dimension. The SAR platform moves along a linear trajectory with an along-track velocity v, where M receiving channels with the adjacent channel distance of d are uniformly installed along the along-track motion direction. The ground maneuvering target is assumed to move from the initial position P_a to the position P_b with the azimuth velocity and acceleration of (v_y, a_y) and the ground cross-track velocity and acceleration of (v_x, a_x). When $t_m = 0$, the positions of the target and SAR platform are supposed to be located at $P_a = (x_0, y_0, 0)$ and $A = (0, 0, H)$, respectively.

According to Fig. 1, assume that the position of the mth receiving channel is $A_m = (0, d_m, H)$ and $d_m = (m-1)d$. Then the equivalent bidirectional slant range between this target and the mth receiving SAR channel can be expressed as follows:

$$\begin{aligned} &R_{s,m}(t_m) \\ &= \frac{1}{2}\left\|\overrightarrow{AP_a} + \overrightarrow{P_aP_b} - \vec{V}t_m\right\| + \frac{1}{2}\left\|\overrightarrow{A_mP_a} + \overrightarrow{P_aP_b} - \vec{V}t_m\right\| \\ &= \frac{1}{2}\sqrt{\begin{array}{c}\left\|\overrightarrow{AP_a}\right\|^2 + \left\|\overrightarrow{P_aP_b}\right\|^2 + \left\|\vec{V}\right\|^2 t_m^2 + 2\overrightarrow{AP_a}\cdot\overrightarrow{P_aP_b} \\ -2\overrightarrow{AP_a}\cdot\vec{V}t_m - 2\overrightarrow{P_aP_b}\cdot\vec{V}t_m\end{array}} \\ &\quad + \frac{1}{2}\sqrt{\begin{array}{c}\left\|\overrightarrow{A_mP_a}\right\|^2 + \left\|\overrightarrow{P_aP_b}\right\|^2 + \left\|\vec{V}\right\|^2 t_m^2 + 2\overrightarrow{A_mP_a}\cdot\overrightarrow{P_aP_b} \\ -2\overrightarrow{A_mP_a}\cdot\vec{V}t_m - 2\overrightarrow{P_aP_b}\cdot\vec{V}t_m\end{array}} \end{aligned} \tag{1}$$

where t_m corresponds to the slow-time variable. $\vec{V} = [0, v, 0]^{\mathrm{T}}$ is the velocity vector of this multichannel SAR platform. $\overrightarrow{P_aP_b} = [v_x t_m + (1/2)a_x t_m^2, v_y t_m + (1/2)a_y t_m^2, 0]^{\mathrm{T}}$ represents the target position vector at the sampling time t_m. $\overrightarrow{AP_a} = [x_0, y_0, -H]^{\mathrm{T}}$ and $\overrightarrow{A_mP_a} = [x_0, y_0 - d_m, -H]^{\mathrm{T}}$ correspond to the position vectors between the initial positions of moving target and the reference channel and between the initial positions of moving target and the mth receiving channel, respectively. Then, $R_{s,m}(t_m)$ in (1) can be rewritten as follows:

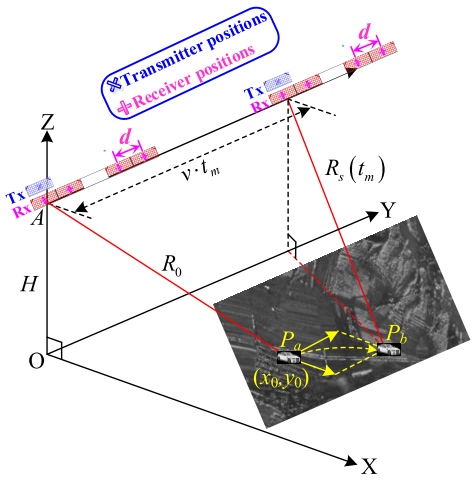


Fig. 1. Three-dimensional observation configuration for a ground moving target in a multichannel SAR system.

$$\begin{aligned} &R_{s,m}(t_m) \\ &= \frac{1}{2}\sqrt{\begin{array}{c}x_0^2 + y_0^2 + H^2 + \left(v_x t_m + \frac{1}{2}a_x t_m^2\right)^2 \\ + \left(v_y t_m + \frac{1}{2}a_y t_m^2\right)^2 + v^2 t_m^2 \\ + 2x_0\left(v_x t_m + \frac{1}{2}a_x t_m^2\right) + 2y_0\left(v_y t_m + \frac{1}{2}a_y t_m^2\right) \\ - 2y_0 v t_m - 2v v_y t_m^2 - v a_y t_m^3\end{array}} \\ &\quad + \frac{1}{2}\sqrt{\begin{array}{c}x_0^2 + (y_0 - d_m)^2 + H^2 \\ + \left(v_x t_m + \frac{1}{2}a_x t_m^2\right)^2 + \left(v_y t_m + \frac{1}{2}a_y t_m^2\right)^2 \\ + v^2 t_m^2 + 2x_0\left(v_x t_m + \frac{1}{2}a_x t_m^2\right) \\ + 2(y_0 - d_m)\left(v_y t_m + \frac{1}{2}a_y t_m^2\right) \\ - 2(y_0 - d_m)v t_m - 2v v_y t_m^2 - v a_y t_m^3.\end{array}} \end{aligned} \tag{2}$$

According to the effective phase center principle in a multichannel SAR system [15], [36], $R_{s,m}(t_m)$ in (2) can be equivalently transformed into the self-transmitting and self-receiving slant range, that is,

$$\begin{aligned} &R_{s,m}(t_m) \\ &\approx \sqrt{\begin{array}{c}x_0^2 + \left(y_0 - \frac{d_m}{2}\right)^2 + H^2 + \left(v_x t_m + \frac{1}{2}a_x t_m^2\right)^2 \\ + \left(v_y t_m + \frac{1}{2}a_y t_m^2\right)^2 + v^2 t_m^2 + 2x_0\left(v_x t_m + \frac{1}{2}a_x t_m^2\right) \\ + 2\left(y_0 - \frac{d_m}{2}\right)\left(v_y t_m + \frac{1}{2}a_y t_m^2\right) \\ - 2\left(y_0 - \frac{d_m}{2}\right)v t_m - 2v v_y t_m^2 - v a_y t_m^3.\end{array}} \end{aligned} \tag{3}$$

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Then, after performing the Taylor series expansion in (3), one has

$$R_{s,m}(t_m) \approx R_0 - \frac{d_m}{2}\cos\theta_{\text{Cone},t} - v\cos\theta_{\text{Cone},t}t_m + v_r t_m + \frac{d_m v}{2R_0}t_m + \frac{v_x^2 + (v - v_y)^2 + x_0 a_x + y_0 a_y}{2R_0}t_m^2 + \frac{a_x v_x + a_y v_y - v a_y}{2R_0}t_m^3 \tag{4}$$

where $R_0 = (x_0^2 + y_0^2 + H^2)^{1/2}$ is the nearest slant range, $v_r = (x_0 v_x + y_0 v_y)/R_0$ represents the target RLOS velocity, $\cos(\theta_{\text{Cone},t}) = y_0/R_0$ is the target cone angle, and $d_m v t_m/2R_0$ indicates the registration error between the mth channel and the reference channel.

Supposing that the multipulse linear frequency modulation (LFM) waveform is used as the transmit signal, then

$$p(t) = \text{rect}\left(\frac{t}{T_p}\right)\exp\left(j\pi\mu t^2\right)\exp(j2\pi f_c t) \tag{5}$$

where

$$\text{rect}\left(\frac{t}{T_p}\right) = \begin{cases} 1, & |t| \le T_p/2 \\ 0, & |t| > T_p/2. \end{cases}$$

The symbols t, μ, f_c, and T_p denote the fast-time variable, the chirp rate of the transmitted signal, the carrier frequency, and the pulsewidth, respectively. Then according to SPP, the received baseband target signal of the mth receiving channel in the range-compression domain can be presented as follows [74]:

$$\begin{aligned} s_{t,m}(t, t_m) &= \sigma_1 \text{sinc}\left\{B\left[t - \frac{2R_{s,m}(t_m)}{c}\right]\right\}\exp\left[-j\frac{4\pi}{\lambda}R_{s,m}(t_m)\right] \\ &= \sigma_1 \text{sinc}\left\{B\left[t - \frac{2R_{s,m}(t_m)}{c}\right]\right\} \\ &\quad \times \exp\left[-j\frac{4\pi}{\lambda}\left(R_0 - \frac{d_m}{2}\cos\theta_{\text{Cone},t}\right)\right] \\ &\quad \times \exp\left\{-j\frac{4\pi}{\lambda}\begin{bmatrix} -v\cos\theta_{\text{Cone},t}t_m + v_r t_m \\ + \dfrac{v_x^2 + (v - v_y)^2 + x_0 a_x + y_0 a_y}{2R_0}t_m^2 \\ [.3cm] + \dfrac{a_x v_x + a_y v_y - v a_y}{2R_0}t_m^3 \end{bmatrix}\right\} \end{aligned} \tag{6}$$

where σ_1, B, λ, c, and $\text{sinc}(x) = \sin(\pi x)/\pi x$ denote, respectively, the target signal complex amplitude in the range-compressed domain, the range bandwidth, the wavelength of the transmitted LFM signal, the speed of light, and the sinc function. The image registration error term in (4) is omitted since it can be precisely compensated by using the spatial correlation method [37]. After implementing the azimuth compression based on the priori radar system parameters, this ground target will be typically smeared in the range-Doppler domain, i.e.,

$$\begin{aligned} s_{t,m}(t, f_a) &= \sigma_{t0}\text{sinc}\left\{B\left[t - \frac{2R_{s,m}(f_a)}{c}\right]\right\} \\ &\quad \times \exp\left(j\frac{\pi d_m f_a}{v}\right)\exp\left(j2\pi\frac{d_m v_r}{\lambda v}\right) \\ &\quad \times \text{sinc}\left[T_a\left(f_a - \frac{2v}{\lambda}\theta_{\text{Cone},t} + \frac{2v_r}{\lambda}\right)\right] \\ &\quad \otimes \text{FFT}_{t_m}\left\{\begin{array}{c}\exp\left[-j\dfrac{4\pi}{\lambda}\left(\dfrac{v_x^2 + v_y^2 - 2vv_y + x_0 a_x + y_0 a_y}{2R_0}t_m^2\right)\right] \\ \times \exp\left[-j\dfrac{4\pi}{\lambda}\left(\dfrac{a_x v_x + a_y v_y - v a_y}{2R_0}t_m^3\right)\right]\end{array}\right\} \end{aligned} \tag{7}$$

where σ_{t0}, f_a, $\otimes$, FFT_{t_m}, and T_a denote the target signal amplitude in t–f_a domain, Doppler frequency, convolution operation, azimuth fast Fourier transform (FFT) operation, and observation time, respectively. It is evident, however, that the ground target energy will be severely dispersed after range-azimuth compression, as the target imaging function is mismatched in virtue of its unknown motions, especially during the relatively long aperture time. In addition, the target true position will be shifted because of v_r.

Similar to the deduction procedure in (7), for a ground clutter scatterer, it can be well focused after 2-D compression in the case of along- and cross-track Doppler parameters being zero, with the focused expression given by

$$\begin{aligned} s_{c,m}(t, f_a) &= \sigma_{c0}\text{sinc}\left\{B\left[t - \frac{2R_{c0,m}}{c}\right]\right\} \\ &\quad \times \exp\left(j\frac{\pi d_m f_a}{v}\right)\text{sinc}\left[T_a\left(f_a - \frac{2v}{\lambda}\theta_{\text{Cone},c}\right)\right] \end{aligned} \tag{8}$$

where σ_{c0}, $R_{c0,m}$, and $\theta_{\text{Cone},c}$ denote the clutter amplitude in t–f_a domain, the scatterer initial distance, and the clutter spatial cone, respectively.

Comparing (7) with (8), one can see that, after implementing the terrain interferometric phase compensation [7], [37], the target spatial steering vector and the clutter spatial steering vector can be, respectively, expressed as follows:

$$\boldsymbol{a}_s = \left[1, \exp\left(\frac{j2\pi d v_r}{\lambda v}\right), \ldots, \exp\left(\frac{j2\pi(M-1)d v_r}{\lambda v}\right)\right]^{\text{T}} \in \mathbb{C}^{M\times 1} \tag{9}$$

$$\boldsymbol{a}_c = [1, 1, \ldots, 1]^{\text{T}} \in \mathbb{C}^{M\times 1} \tag{10}$$

where "T" represents the transpose operation. Due to the differences in target and clutter spatial vectors, the troubled clutter signal can be effectively suppressed by fully utilizing the precious space–time degree of freedom in a multichannel radar system, where the clutter filtering can be well accomplished by using the well-known post-Doppler domain-based space–time adaptive processing (STAP) approaches in [27], [28], [29], [30], [31], and [32] or the SAR image-domain-based adaptive filtering methods in [33], [34], [35], [36], and [37].

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After implementing the adaptive filtering processing for the unwanted clutter components, the target signal in (7) will be coherently synthetized along spatial domain, and thus its SCNR will be greatly improved. Then, after implementing the inverse range-azimuth compression procedure on the spatially synthetized target signal, the smeared target signal in f_r–t_m domain can be presented as follows:

$$s_{t1}(f_r,t_m)\approx\sigma_2\mathrm{rect}\left(\frac{f_r}{B}\right)\exp\left[-j\frac{4\pi}{c}f_r\left(R_0+c_1t_m+c_2t_m^2\right)\right]\times\exp\left[-j\frac{4\pi}{\lambda}\left(R_0+c_1t_m+c_2t_m^2+c_3t_m^3\right)\right] \quad (11)$$

where f_r represents the frequency variable corresponding to t, and σ_2 represents the target amplitude in f_r–t_m domain after performing the spatial synthesis. The symbols c_1, c_2, and c_3 correspond to the first-, second-, and third-order Doppler coefficients, respectively, which are mainly determined by the target along- and cross-track motions, i.e.,

$$\begin{cases} c_1=-v\cos\theta_{\mathrm{Cone},t}+v_r \\ c_2=\dfrac{v_x^2+\left(v-v_y\right)^2+x_0a_x+y_0a_y}{2R_0} \\ c_3=\dfrac{a_xv_x+a_yv_y-va_y}{2R_0}. \end{cases} \quad (12)$$

It is evident from (11) that the coupling effect between f_r and t_m will cause the center position of target envelope to vary with different pulse sampling moments, i.e., the troubled RCM occurs. It is notable that the target third-order RCM is ignored in (11) since the third-order envelope shift is usually smaller than the SAR resolution [53], [54], [55], [56], [57], [58], [59]. Moreover, the complex azimuth phase terms related to the target unknown parameters will introduce the asymmetrical Doppler broadening, causing the considerable SCNR loss if they are not properly handled. Therefore, to enhance the subsequent target detection and relocation performances, the phase coupling terms induced by the target unknown motions should be accurately removed.

III. Proposed Algorithm Description

A. Ground Maneuvering Target Imaging

In order to remove the linear RCM in (11), KT [51], [52] is exploited to realize the linear phase decoupling via the time-scaled transform. Then one has

$$s_2(f_r,\tau_m)=\sigma_2\mathrm{rect}\left(\frac{f_r}{B}\right)\exp\left[-j\frac{4\pi f_r}{c}\left(R_0-c_2\tau_m^2\right)\right]\times\exp\left[-j\frac{4\pi}{\lambda}\left(R_0+c_1\tau_m+c_2\tau_m^2+c_3\tau_m^3\right)\right] \quad (13)$$

where $\tau_m=m\cdot P_{\mathrm{RT}}$ represents the new slow-time variable with P_{RT} being the pulse sampling time. The above derivation procedure applies the approximations of $f_c/(f_r+f_c)=1-f_r/f_c$ and $f_c^2/(f_r+f_c)^2\approx1-2f_r/f_c$ [55].

After removing the adverse influence introduced by the target first-order term, the range curvature is still present. Motivated by the previously proposed time reversion transform (TRT) in [54] and range frequency reversion transform (RFRT) in [15], the MSF is constructed to completely remove the defocusing effect caused by c_2, that is,

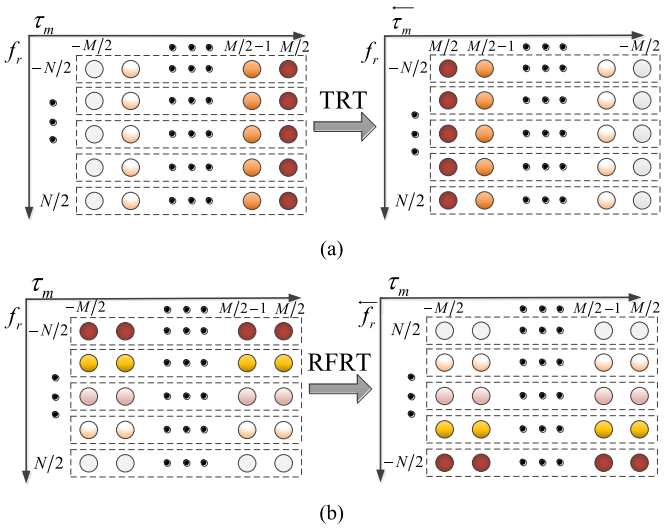


Fig. 2. Schematic of the TRT and RFRT operations. Data arrangement scheme by using (a) TRT and (b) RFRT.

$$s_{\mathrm{MSF}}(f_r,\tau_m)=s_{\mathrm{TR}}\left(f_r,\overleftarrow{\tau_m}\right)\cdot s_{\mathrm{FR}}\left(\overleftarrow{f_r},\tau_m\right) \quad (14)$$

where the symbol "$\leftarrow$" above τ_m and "f_r" denotes the TRT and RFRT operations, respectively, as shown in Fig. 2. Based on (13) and Fig. 2, the expressions of TRT and RFRT are given by

$$s_{\mathrm{TRT}}\left(f_r,\overleftarrow{\tau_m}\right)=\sigma_2\mathrm{rect}\left(\frac{f_r}{B}\right)\exp\left[-j\frac{4\pi f_r}{c}\left(R_0-c_2\tau_m^2\right)\right]\times\exp\left[-j\frac{4\pi}{\lambda}\left(R_0-c_1\tau_m+c_2\tau_m^2-c_3\tau_m^3\right)\right] \quad (15)$$

$$s_{\mathrm{RFRT}}\left(\overleftarrow{f_r},\tau_m\right)=\sigma_2\mathrm{rect}\left(\frac{f_r}{B}\right)\exp\left[j\frac{4\pi f_r}{c}\left(R_0-c_2\tau_m^2\right)\right]\times\exp\left[-j\frac{4\pi}{\lambda}\left(R_0+c_1\tau_m+c_2\tau_m^2+c_3\tau_m^3\right)\right]. \quad (16)$$

Substituting (15) and (16) into (14), one has

$$s_{\mathrm{MSF}}(f_r,\tau_m)=\sigma_2^2\mathrm{rect}\left(\frac{f_r}{B}\right)\exp\left[-j\frac{8\pi}{\lambda}\left(R_0+c_2\tau_m^2\right)\right]. \quad (17)$$

Obviously, from (17), one can see that the RCM is completely eliminated and Doppler phase terms corresponding to the target equivalent first- and third-order terms are removed. After implementing the range inverse FFT (IFFT) on (17), we obtain

$$s_{\mathrm{MSF}}(t,\tau_m)=\sigma_3\mathrm{sinc}(Bt)\exp\left[-j\frac{8\pi}{\lambda}\left(R_0+c_2\tau_m^2\right)\right] \quad (18)$$

where σ_3 is the target complex amplitude in t–τ_m domain. At last, after employing the NUFFT [79], [80] on (18) with respect to τ_m^2, a ground target can be efficiently accumulated

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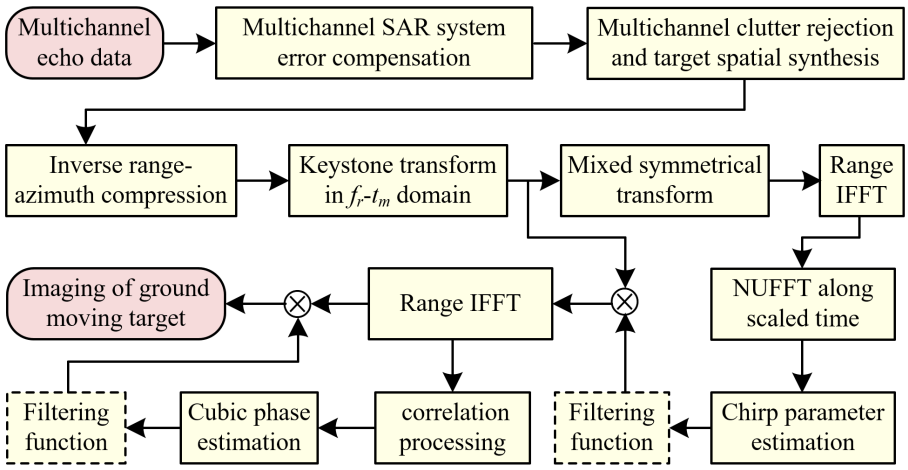


Fig. 3. Ground target imaging by using the proposed method.

into a well-integrated peak in the scaled-frequency domain, i.e.,

$$s_{\mathrm{MSF}}\left(t, f_{\tau_m^2}\right) = \mathrm{NUFFT}_{\tau_m^2}[s_{\mathrm{MSF}}(t, \tau_m)] = \sigma_4 \mathrm{sinc}(Bt)\mathrm{sinc}\left[T_a\left(f_{\tau_m^2} + \frac{4}{\lambda}c_2\right)\right] \tag{19}$$

where $\mathrm{NUFFT}_{\tau_m^2}[\cdot]$ denotes the NUFFT operation corresponding to τ_m^2, σ_4 is the target complex amplitude in $t-f_{\tau_m^2}$ domain, and $f_{\tau_m^2}$ stands for the scaled frequency corresponding to τ_m^2. At the same time, the target equivalent second-order coefficient can be retrieved as follows:

$$\hat{c}_2 = -\frac{\lambda}{4}\arg\max_{f_{\tau_m^2}}\left|s_{\mathrm{MSF}}\left(t, f_{\tau_m^2}\right)\right|. \tag{20}$$

In the following, we construct a compensation function to eliminate the coupling effect induced by c_2 in (13), that is,

$$H_{c_2}(f_r, \tau_m) = \exp\left(-j\frac{4\pi f_r}{c}c_2\tau_m^2\right)\exp\left(j\frac{4\pi}{\lambda}c_2\tau_m^2\right). \tag{21}$$

Then after implementing the IFFT along range dimension, one has

$$s_3(t, \tau_m) = \sigma_3 \mathrm{sinc}\left[B\left(t - \frac{2R_0}{c}\right)\right] \times \exp\left[-j\frac{4\pi}{\lambda}\left(R_0 + c_1\tau_m + c_3\tau_m^3\right)\right]. \tag{22}$$

It is apparently seen from (22) that the RCM is effectively removed and the target signal can be described as a CPS with the absence of second-order phase term. After performing the following correlation operation, the CPS term can be reduced to the second-order one, that is,

$$s_4(\tau_m) = s_3\left(\frac{2R_0}{c}, \tau_m + \frac{\tau_0}{2}\right)s_3^*\left(\frac{2R_0}{c}, \tau_m - \frac{\tau_0}{2}\right) = \sigma_4 \exp\left[-j\frac{4\pi}{\lambda}\left(c_1\tau_0 + \frac{1}{4}c_3\tau_0^3 + 3\tau_0 c_3\tau_m^2\right)\right] \tag{23}$$

where τ_0 denotes the fixed lag-time constant, which can be set as $T_a/4$ empirically [55]. As it is apparent from (23), after employing the NUFFT along the τ_m^2-axis, this second-order PPS generated in the scaled-frequency domain is given as follows:

$$s_4\left(f_{\tau_m^2}\right) = \mathrm{NUFFT}_{\tau_m^2}[s_4(\tau_m)] = \sigma_5 \mathrm{sinc}\left[T_a\left(f_{\tau_m^2} + \frac{6}{\lambda}\tau_0 c_3\right)\right]. \tag{24}$$

Then, based on the focused result in (24), c_3 can be efficiently estimated as follows:

$$\hat{c}_3 = -\frac{\lambda}{6\tau_0}\arg\max_{f_{\tau_m^2}}\left|s_4\left(f_{\tau_m^2}\right)\right|. \tag{25}$$

Finally, based on $\hat{c}_2$ and $\hat{c}_3$, the target signal can be well integrated in t–f_a domain, with the expression given by

$$s_4(t, f_a) = \sigma_6 \mathrm{sinc}\left[B\left(t - \frac{2R_0}{c}\right)\right]\mathrm{sinc}\left[T_a\left(f_a + \frac{2c_1}{\lambda}\right)\right] \tag{26}$$

where σ_6 is the integrated target signal amplitude in t–f_a domain. The flowchart of the proposed approach is depicted in Fig. 3, where the multichannel SAR system error compensation procedures, such as multichannel SAR image registration, terrain interference phase compensation, and channel amplitude/phase error compensation, should be considered. Considering that the main theme of this article focuses on the target Doppler parameter estimation and imaging, these multichannel SAR error compensation procedures are not discussed in detail. The interested reader is referred to [37] for more details.

B. Performance Analysis of the Proposed Approach

1) Computational Cost Analysis: Let the numbers of range bins and transmitted radar pulses be N and M_0, respectively. The searching points of c_2 and c_3 are denoted by M_{c_2} and M_{c_3}, respectively.

As for the 2-D frequency matched filtering (MF) approach in [75], it accomplishes the chirp estimation and target imaging in f_r–f_a domain based on a 1-D searching. Hence, its computational load is about $M_{c_2} \cdot O(NM_0\log_2 M_0 + NM_0\log_2 N)$, as listed in Table I, which is relatively efficient with the total

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TABLE I
COMPUTATIONAL COMPLEXITY COMPARISON

Methods	Computational complexities
Proposed method	$O(4NM_0\log_2 M_0)+5NM_0+2\cdot O(M_0\log_2 M_0)$
2-D frequency matched filtering method	$M_{c_2}\cdot O(NM_0\log_2 M_0+NM_0\log_2 N)$
Time reversion transform method	$O(4NM_0\log_2 M_0)+6NM_0$
KT-matched filtering method	$NM_{c_2}M_{c_3}O(M_0\log_2 M_0)$ $+O(4NM_0\log_2 M_0)+4NM_0$

computational burden about $N^3O(\log_2 N)$ in the case of $N = M_0 = M_{c_2}$. As an efficient target imaging method, the TRT was proposed in [54], which can realize the ground uniformly moving target refocusing without any search. Its computational complexity is about $O(4NM_0\log_2 M_0) + 6NM_0$, which is much more efficient with the total computational load about $N^2O(\log_2 N)$ in the case of $N = M_0$. However, when the ground target undergoes the complicated maneuvering motions, the smearing influence caused by the high-order phase component should not be ignored. Then the above two methods may encounter the severe SCNR loss issue.

As an optimal approach based on the third-order motion model, the KT-MF (KTMF) method in [81] realizes the maneuvering target imaging by the 2-D grid searching. Its time-consuming procedures mainly include the KT according to scaling principle [52], [53] with the computational load of $O(4NM_0\log_2 M_0) + 4NM_0$ and a 2-D searching of target Doppler parameters ($NM_{c_2}M_{c_3}O(M_0\log_2 M_0)$). Because of the exhaustive computationally demanding search with respect to the filter bank containing lots of combinations of the phase components, this method is computationally demanding with the total computational load about $N^4O(\log_2 N)$ in the case of $N = M_0 = M_{c_2} = M_{c_3}$.

The main time-consuming steps of the proposed approach include the KT ($O(4NM_0\log_2 M_0) + 4NM_0$), the MSF with the computational load of NM_0, and the second- and third-order parameter estimations based on two times of NUFFTs, where the NUFFT procedure with the computational load of $O(M_0\log_2 M_0)$ is much more similar to the well-known FFT [79], [80]. Therefore, the total computational cost is about $O(4NM_0\log_2 M_0)+5NM_0+2\cdot O(M_0\log_2 M_0)$, which is approximately $N^2O(\log_2 N)$ in the case of $N = M_0$. Obviously, the proposed approach is also efficient to accomplish the ground maneuvering target parameter estimation and imaging since it does not require the grid search operations.

In the following, the practical running times by the above-mentioned approaches are compared via the Monte Carlo experiments, where the range gate number is set as 144, and the searching points of c_2 and c_3 are both set as 144 for convenience. For each given azimuth pulse number, 100 Monte Carlo simulations are provided, and the varying relationships between the running time and azimuth pulse number are depicted in Fig. 4 (the processor of the computer is Intel[1] Xeon[1] Gold 6226R CPU at 2.90 and 2.89 GHz, and the RAM is 64 GB). From the figure, we see that the proposed approach is computationally efficient similar to the TRT approach, but consuming much less time than those of the 2-D frequency MF approach and the KTMF approach.

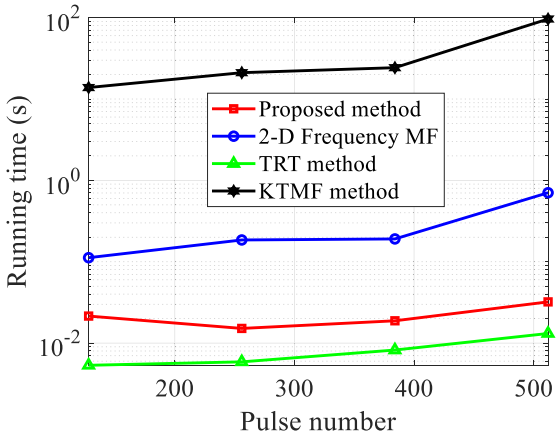


Fig. 4. Relationships between the running time and the pulse number, where the range gate number, the searching point of c_2, and the searching point of c_3 are set as 144 (the processor of the computer is Intel Xeon Gold 6226R CPU at 2.90 and 2.89 GHz, and the RAM is 64 GB).

2) Multitarget Focusing Analysis: Usually, ground targets will be located at different range-Doppler positions in an SAR image, which can be well focused one by one. However, when two targets are located at the close range and Doppler positions, the cross-terms will unavoidably be introduced by the proposed method due to the nonlinear transform. According to (11), assuming that the expression for the multiple moving targets in f_r–t_m domain is given by

$$s_{t,\mathrm{Mul}}(f_r,t_m)=\sum_{i=1}^{K}\sigma_{ti}\exp\left[-j\frac{4\pi}{c}f_r\left(R_{0,i}+c_{1,i}t_m+c_{2,i}t_m^2\right)\right]\times\exp\left[-j\frac{4\pi}{\lambda}\left(R_{0,i}+c_{1,i}t_m+c_{2,i}t_m^2+c_{3,i}t_m^3\right)\right] \tag{27}$$

where K is the number of moving targets. The symbols σ_{ti}, $R_{0,i}$, $c_{1,i}$, $c_{2,i}$, and $c_{3,i}$ represent, respectively, the signal amplitude, the initial range, the first-order, the second-order, and the third-order coefficients of the ith target. Usually, because of different range-azimuth positions as well as along- and cross-track motions, the polynomial coefficients of these targets are usually different.

After implementing the KT and the MSF transform in (27), one has

$$s_{\mathrm{MSF,Mul}}(f_r,\tau_m)=s_{\mathrm{MSF,auto}}(f_r,\tau_m)+s_{\mathrm{MSF,cross}}(f_r,\tau_m) \tag{28}$$

where $s_{\mathrm{MSF,auto}}(f_r,\tau_m)$ represents the autoterms and $s_{\mathrm{MSF,cross}}(f_r,\tau_m)$ denotes the cross-terms caused by the nonlinear operations in the MSF transform, i.e.,

$$s_{\mathrm{MSF,auto}}(f_r,\tau_m)=\sum_{i=1}^{K}\sigma_{ti}\mathrm{rect}\left(\frac{f_r}{B}\right)\times\exp\left[-j\frac{4\pi}{\lambda}\left(2R_{0,i}+2c_{2,i}\tau_m^2\right)\right] \tag{29}$$

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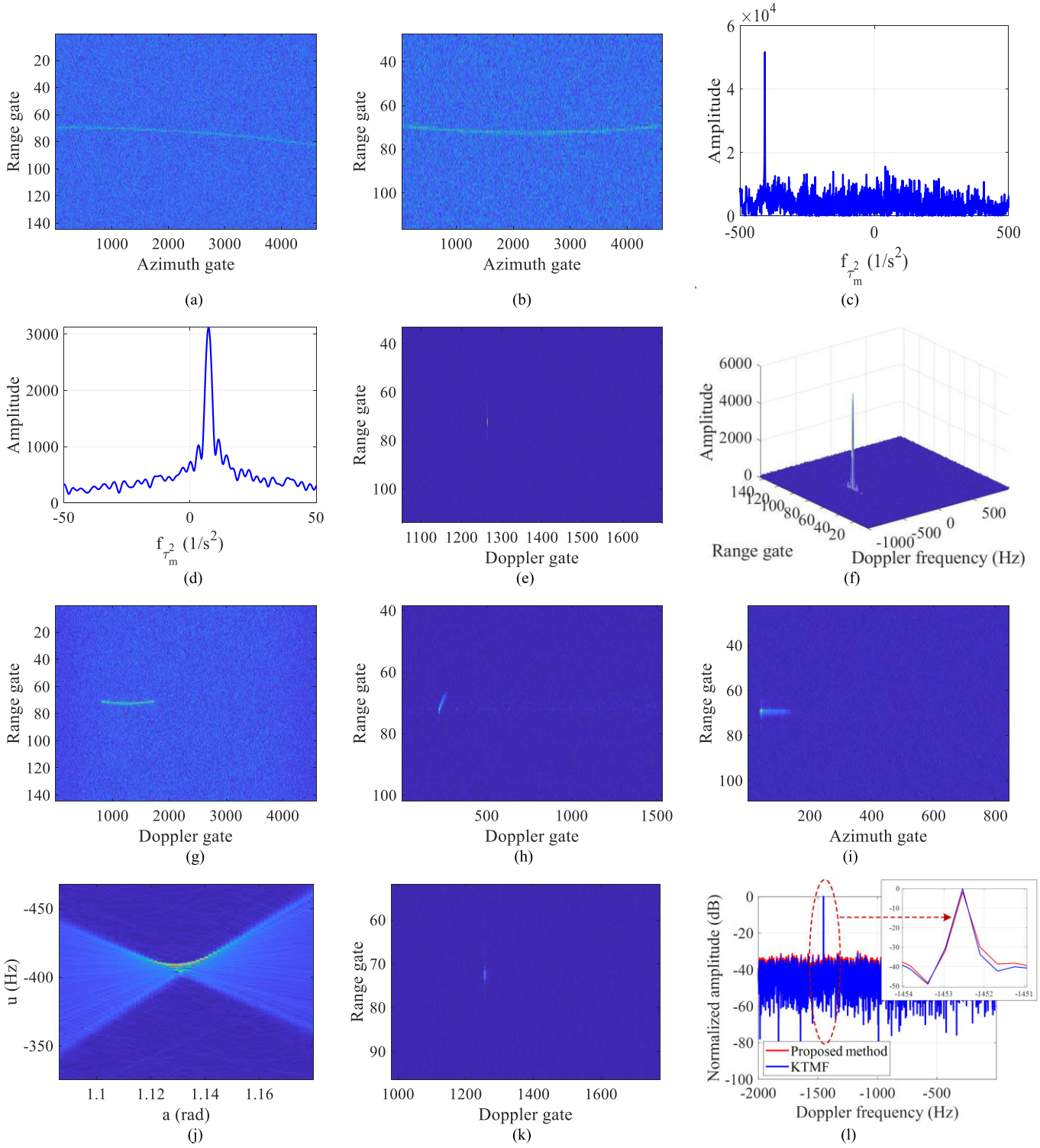


Fig. 5. Simulated target focusing results with the range-compressed SNR being 5 dB. (a) Range compression. (b) KT result. (c) NUFFT result with respect to (18). (d) NUFFT result with respect to (23). (e) Imaging result by the proposed method. (f) Stereogram of focusing result of (e). Imaging result by (g) DKT method, (h) TRT method, (i) 2-D frequency MF method, (j) Radon-FrFT method, and (k) KTMF method. (l) Comparison results between the KTMF method and the proposed method.

$$
\begin{aligned}
&s_{\mathrm{MSF,cross}}(f_r,\tau_m)\\
&=\sum_{i=1}^{K}\sum_{j=1\,i\neq j}^{K}\sigma_{ti}\sigma_{tj}\mathrm{rect}\left(\frac{f_r}{B}\right)\\
&\quad\times\exp\left[-j\frac{4\pi}{c}f_r\left(R_{0,i}-c_{2,i}\tau_m^2-R_{0,j}+c_{2,j}\tau_m^2\right)\right]\\
&\quad\times\exp\left[-j\frac{4\pi}{\lambda}\left(\begin{array}{c}R_{0,i}+R_{0,j}-c_{1,i}\tau_m+c_{1,j}\tau_m+c_{2,i}\tau_m^2\\+c_{2,j}\tau_m^2-c_{3,i}\tau_m^3+c_{3,j}\tau_m^3\end{array}\right)\right].
\end{aligned}
\tag{30}
$$

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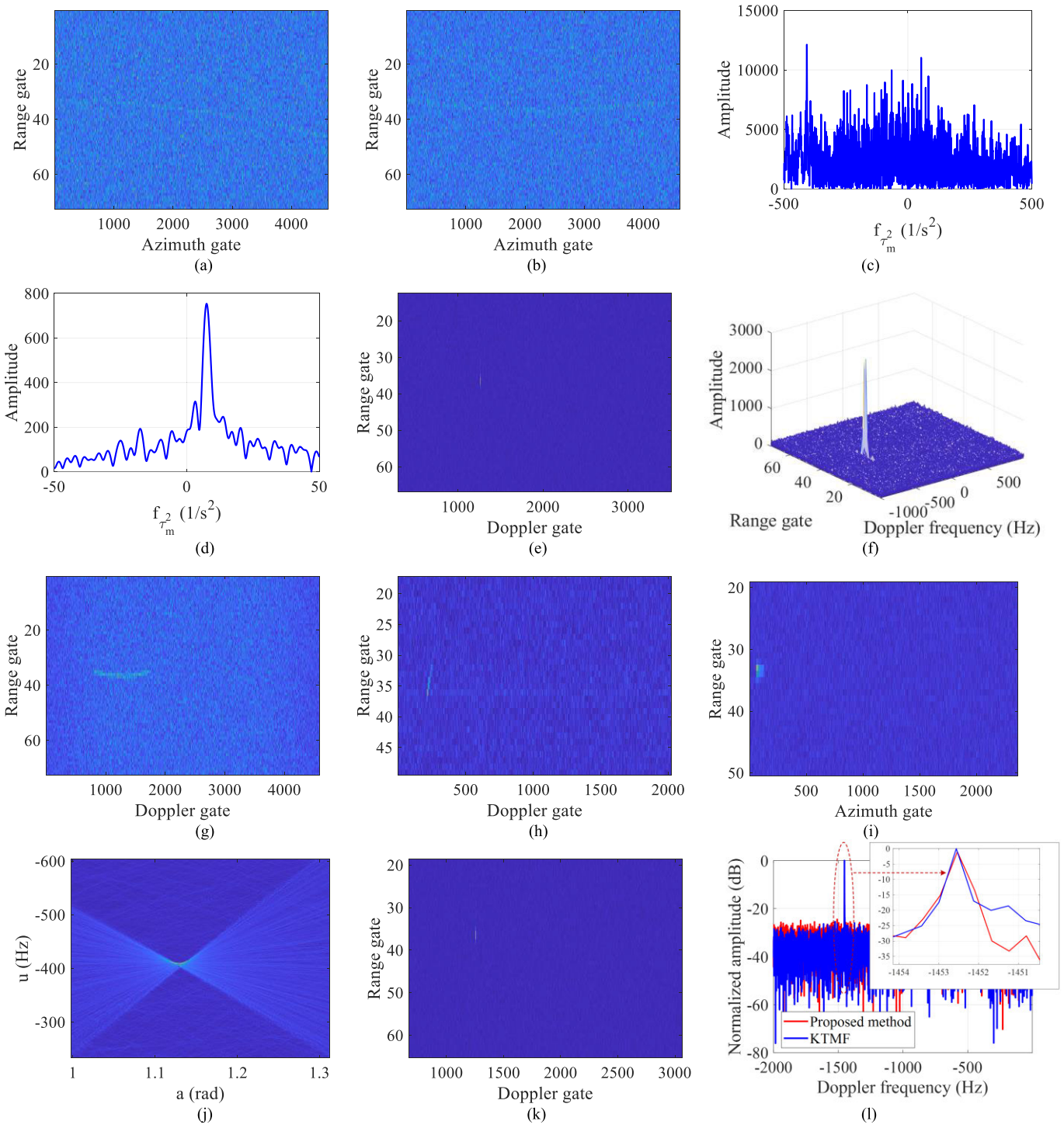


Fig. 6. Simulated target focusing results when the range-compressed SNR of a ground moving target is −2 dB. (a) Range compression. (b) KT result. (c) NUFFT result with respect to (18). (d) NUFFT result with respect to (23). (e) Imaging result by the proposed method. (f) Stereogram of focusing result of (e). Imaging result by (g) DKT method, (h) TRT method, (i) 2-D frequency MF method, (j) Radon-FrFT method, and (k) KTMF method. (l) Comparison results between the KTMF method and the proposed method.

After implementing the range IFFT in (29) and (30), we have

$$s_{\mathrm{MSF,auto}}(t,\tau_m) = \sum_{i=1}^{K}\sigma_{ti,\mathrm{auto}}\mathrm{sinc}(Bt)\times\exp\left[-j\frac{4\pi}{\lambda}\left(2R_{0,i}+2c_{2,i}\tau_m^2\right)\right] \quad (31)$$

$$s_{\mathrm{MSF,cross}}(t,\tau_m) = \sum_{i=1}^{K}\sum_{\substack{j=1\\ i\neq j}}^{K}\sigma_{t,i,j,\mathrm{cross}} \times \mathrm{sinc}\left\{B\left[t-\frac{2}{c}\left(R_{0,i}-c_{2,i}\tau_m^2-R_{0,j}+c_{2,j}\tau_m^2\right)\right]\right\}$$

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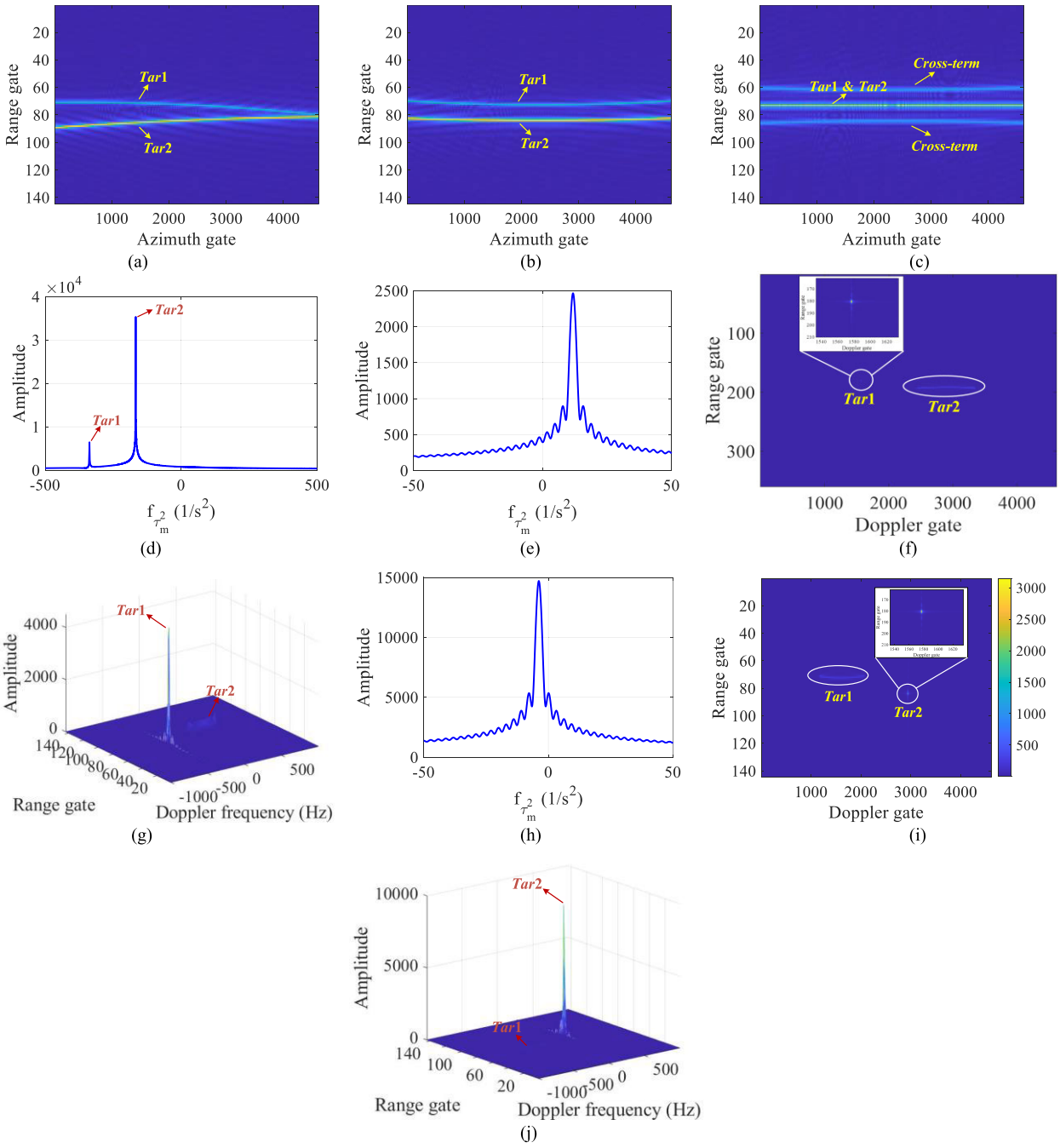


Fig. 7. Two simulated target imaging results. (a) Range compression. (b) Target trajectories after KT. (c) Range profiles after MSF transform. (d) NUFFT result with respect to (18). (e) NUFFT result of *Tar*1 with respect to (23). (f) Imaging result of *Tar*1. (g) Stereogram of focusing result of *Tar*1. (h) NUFFT result of *Tar*2 with respect to (23). (i) Imaging result of *Tar*2. (j) Stereogram of focusing result of *Tar*2.

$$\times \exp\left[-j\frac{4\pi}{\lambda}\left(\begin{array}{c} R_{0,i}+R_{0,j}-c_{1,i}\tau_m+c_{1,j}\tau_m+c_{2,i}\tau_m^2 \\ +c_{2,j}\tau_m^2-c_{3,i}\tau_m^3+c_{3,j}\tau_m^3 \end{array}\right)\right] \tag{32}$$

where $\sigma_{ti,\text{auto}}$ and $\sigma_{t,i,j,\text{cross}}$ denote, respectively, the autoterm and cross-term amplitudes.

Obviously, we can see from (31) that the autoterms will be located at $t = 0$, and the range migrations of autoterms are effectively removed via the MSF transform. In addition, only the second-order phase terms are reserved, which are beneficial to the subsequent NUFFT processing with respect to τ_m^2.

However, the residual RCM is still present for the cross-terms in (32). Thus, they will not be located at $t = 0$ in the range-compression domain, which may not influence the search of autoterms. Moreover, both τ_m-related and τ_m^3-related phase terms appear, indicating that the cross-terms will be seriously diffused in the scaled-frequency domain (a detailed analysis is similar to those provided in [71], [72], and [73]). In practice, the range-azimuth positions of smeared moving targets can be roughly known as a priori information by applying the CA-constant false alarm rate (CA-CFAR) technique with the 2-D sliding window after implementing

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Fig. 8. Stationary scene SAR image containing a smeared ground moving target and optical image provided by Google map.

the multichannel adaptive clutter rejection. Therefore, the subsequent motion parameter estimation, target refocusing, and relocation can be effectively realized by processing the local region containing the smeared moving targets in the complex multitarget scenarios.

In addition, when multiple targets with different signal intensities coexist, we can utilize the "CLEAN" technique [82] to adaptively alleviate the submerged influence induced by strong targets. Thus, the relatively weak targets can also be well focused.

IV. Simulation and Experiment Analysis Results

A. Simulation Result Analysis

In this part, simulation experiments for a ground maneuvering target are conducted in the complex Gaussian noise background, with the target parameters listed in Table II. Then, according to the radar parameters provided in Table III, c_1, c_2, and c_3 of this target are, respectively, calculated as 7.0711 m/s, 3.1848 m/s^2, and -0.0335 m/s^3, respectively. In this example, the processing results of the deramp-keystone transform (deramp-KT) (DKT) method [14], the keystone transform-time reverse transform (KT-TRT) method [54], the 2-D frequency MF method [75], the Radon-fractional Fourier transform (FrFT) method [78], and the keystone transform matched filtering (KTMF) method [81] will be applied for comparisons.

After range compression, Fig. 5(a) depicts the target trajectory in t–t_m domain, from which we can see that both the linear and second-order RCMs appear. After removing the linear envelope shift by KT, only the second-order RCM exists, as provided in Fig. 5(b). After performing the MSF transform and the NUFFT on the target signal at $t = 0$ [see (18)] along the scaled slow-time dimension, Fig. 5(c) exhibits the focusing result in the scaled-frequency domain, with c_2 estimated as 3.185 m/s^2. After compensating the adverse influence caused by c_2, according to (24), c_3 can be estimated as -0.0332 m/s^3, as provided in Fig. 5(d). Finally, after removing the target motion influence, Fig. 5(e) and (f) displays the target well-imaged result by using the proposed method. Fig. 5(g) displays the target image result by utilizing the DKT method. Although the target energy can be integrated to a certain extent, the target focused peak is still dispersed over multiple range and Doppler units due to the incomplete compensation of the target second-order phase and the neglect of the target third-order phase errors. For the same reason, the smeared results also appear by using the KT-TRT method, the 2-D frequency MF method, and the Radon-FrFT method

TABLE II
Target Motion Parameters

Target along-track velocity	14 m/s
target along-track acceleration	3 m/s^2
target cross-track velocity	10 m/s
target cross-track velocity acceleration	5 m/s^2
range-compressed signal-to-noise ratio (SNR)	5 dB

TABLE III
SAR System Parameters

Wavelength	0.0313 m
Range bandwidth	100 MHz
Range sampling frequency	120 MHz
Pulse repetition frequency	2000 Hz
Platform velocity	120 m/s
Synthetic aperture time	2.3073 s
Pulse during time	1 μs
Target nearest slant range	4 km

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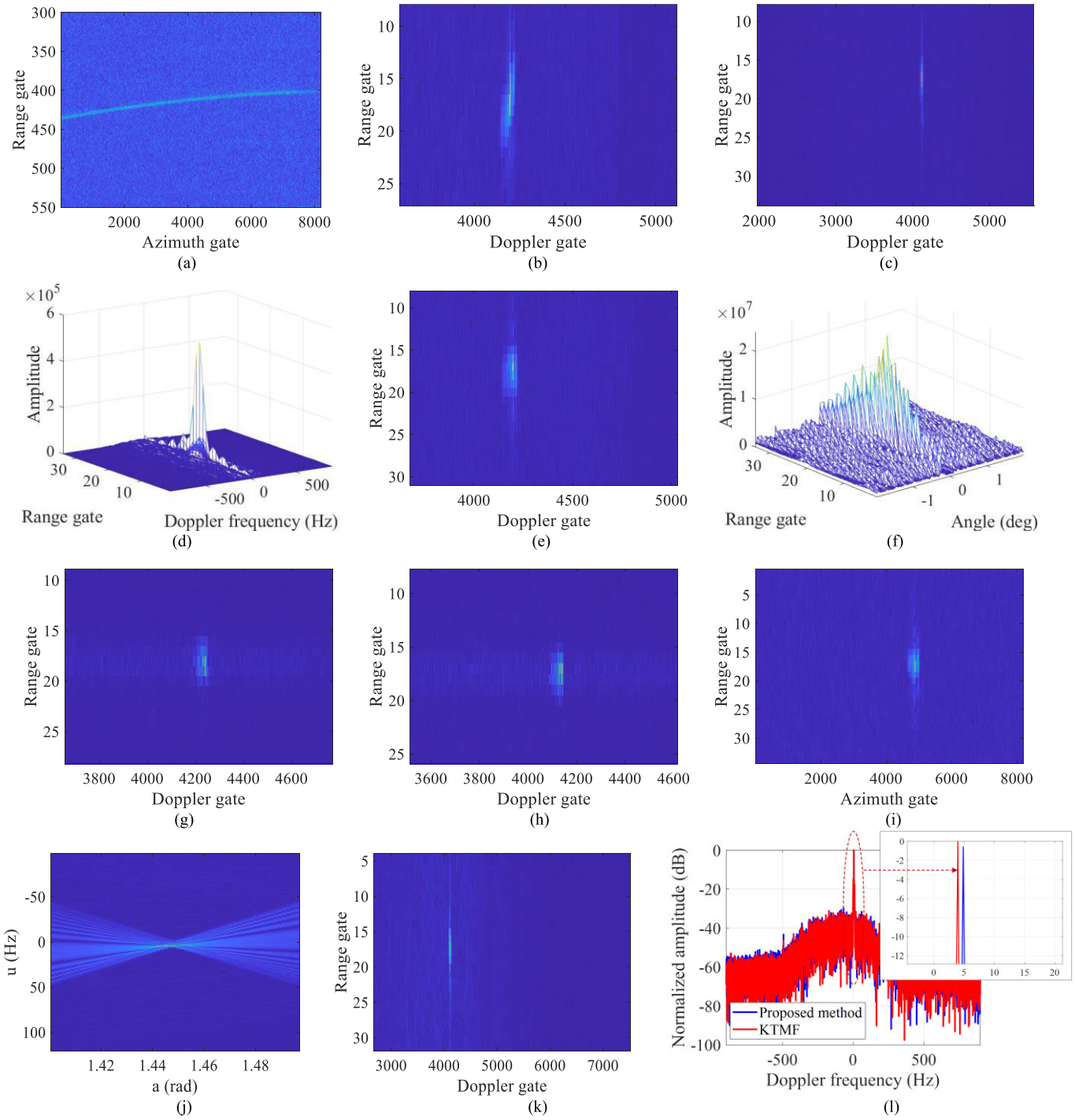


Fig. 9. Dual-channel SAR processing results. (a) Target motion trajectory. (b) Target focusing result by using the multichannel STAP method in [36]. (c) Target imaging result by using the proposed approach. (d) Stereogram of target focusing result with respect to (c). (e) Target imaging result by using the DKT method. (f) Target rotation angle searching result by the IAR-TRT. Target imaging result by using (g) IAR-TRT and (h) KT-TRT method. Imaging result by (i) 2-D frequency MF method, (j) Radon-FrFT approach, and (k) KTMF approach. (l) Compared results by the KTMF approach and the proposed approach.

since the used second-order signal model may not accurately portray the nonstationary phase varying property of this ground maneuvering target, as displayed in Fig. 5(h)–(j). In addition, due to the sufficient compensation of target RCM and Doppler broadening caused by the target complicated motions, this target is finely imaged by using the optimal KTMF method [see Fig. 5(k)], and the azimuth slice is provided in Fig. 5(l). It is obvious from the compared results that the target image quality of the proposed method is similar to that of the KTMF method, but with much less computational complexity.

Fig. 6(a) and (b) depicts the target trajectory in t–t_m domain and the KT processing result, from which we can hardly see the target motion trajectory due to the weak target intensity (the target range-compressed SNR is set as -2 dB). From the

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TABLE IV
TARGET MOTION PARAMETERS

	v_x	a_x	v_y	a_y
*Tar*1	7 m/s	3 m/s^2	8 m/s	4 m/s^2
*Tar*2	-6 m/s	-2 m/s^2	-7 m/s	-1 m/s^2

TABLE V
TARGET PARAMETER ESTIMATION RESULTS

	c_2	c_3
*Tar*1	2.635 m/s^2	-0.053255 m/s^3
*Tar*2	1.3041 m/s^2	0.017305 m/s^3

TABLE VI
REAL-MEASURED SAR SYSTEM PARAMETERS

Pulse repetition frequency	1800 Hz
Wavelength	0.0556 m
Platform velocity	139 m/s
Pulse number	8192
Platform height	8.1 km
Pulse width	10 μs
Squint angle (degree)	0
range resolution	0.5625 m

simulation results in Fig. 6(c)–(f), one can see that this ground weak target is finely focused by using the proposed method. On the contrary, due to the neglect of target third-order phase errors, the target imaging results by using the DKT method, the KT-TRT method, the 2-D frequency MF method, and the Radon-FrFT method are still smeared, as displayed in Fig. 6(g)–(j). In addition, this moving target is also well imaged by using the optimal KTMF method, as depicted in Fig. 6(k) and (l).

Then, two ground targets with the close ranges, denoted by *Tar*1 and *Tar*2, are applied to verify the proposed method. The SAR system parameters are shown in Table III. The target motion parameters are provided in Table IV, where the signal amplitude of *Tar*2 is twice that of *Tar*1. Based on the radar and moving target parameters, c_1, c_2, and c_3 of *Tar*1 are, respectively, calculated as 4.9497 m/s, 2.6348 m/s^2, and -0.053375 m/s^3, and c_1, c_2, and c_3 of *Tar*2 are calculated as -4.2576 m/s, 1.3039 m/s^2, and 0.017314 m/s^3, respectively.

Fig. 7(b) depicts the linear RCM compensation result with respect to Fig. 7(a). Fig. 7(c) gives the MSF transform results after range IFFT. From the figure, one can see that both autoterm trajectories are well aligned and located at the middle range gate. However, the range migration is still present for the cross-terms, which are situated at other range gates. After extracting the autoterm signal data at $t = 0$ and performing the NUFFT with respect to τ_m^2, two well-focused autoterm peaks are generated, as depicted in Fig. 7(d). After compensating the second-order phase error influence, Fig. 7(e) depicts the NUFFT result with respect to (23), with the peak position determining the estimation result of c_3 (see the estimation result in Table IV). Finally, after performing the motion compensation of *Tar*1, Fig. 7(f) and (g) displays the well-imaged results of *Tar*1. It should be noted that *Tar*2 is smeared since the filtering response function related to the Doppler parameters of *Tar*1 will cause the mismatch of the imaging function of *Tar*2. For the same reason, based on the estimation result in Fig. 7(d) and (h), *Tar*2 is also finely imaged, as provided in Fig. 7(i) and (j). For the same reason, when *Tar*2 is fully focused, the energy of *Tar*1 is dispersed among the many range and Doppler gates because of the phase mismatch. Therefore, the proposed approach can effectively deal with the case of two moving targets, with the parameter estimation results provided in Table 5.

B. Real-Measured Airborne SAR Result Analysis

In this section, the high-resolution airborne SAR data recorded by a dual-channel SAR is provided, where the observation scene covers a rural area of Hainan province, China in 2016 (the longitude and latitude of the observations scene of this field experiment are about 109.974° and 18.485°, respectively), and the radar system parameters are listed in Table VI. The SAR imaging result is presented in Fig. 8, where a ground experiment car was driving along a rotary island road of Hainan province, as labeled by ellipses. From the figure, one can see that this moving car is severely smeared due to the phase mismatch between the target imaging function and the ground static scatterer filtering response function. The target imaging results are provided in Fig. 9.



Fig. 10. SAR imaging result with the presence of a smeared ground target.

Fig. 9(a) depicts the target motion trajectory in the slow-time domain after clutter rejection [36], and the visible range position shift and Doppler disperse appear. After performing the target focusing by using the SAR system parameters, this moving target is typically smeared in the SAR image domain due to the target unknown motions, as displayed in Fig. 9(b), thus the target Doppler parameters should be precisely estimated to accomplish the subsequent target focusing. After performing the RCM and Doppler broadening compensation based on high-order parameter estimation, this moving car is finely imaged with a fully integrated peak by the proposed approach, as provided in Fig. 9(c) and (d). Conversely, since the target along-track motions are ignored by the DKT method, this car is smeared in the residual clutter and noise background [see Fig. 9(e)]. In addition, for the same reason, because the third-order phase error is not well dealt with by the IAR-TRT method [9], the KT-TRT method, the 2-D frequency MF method, and the Radon-FrFT method,

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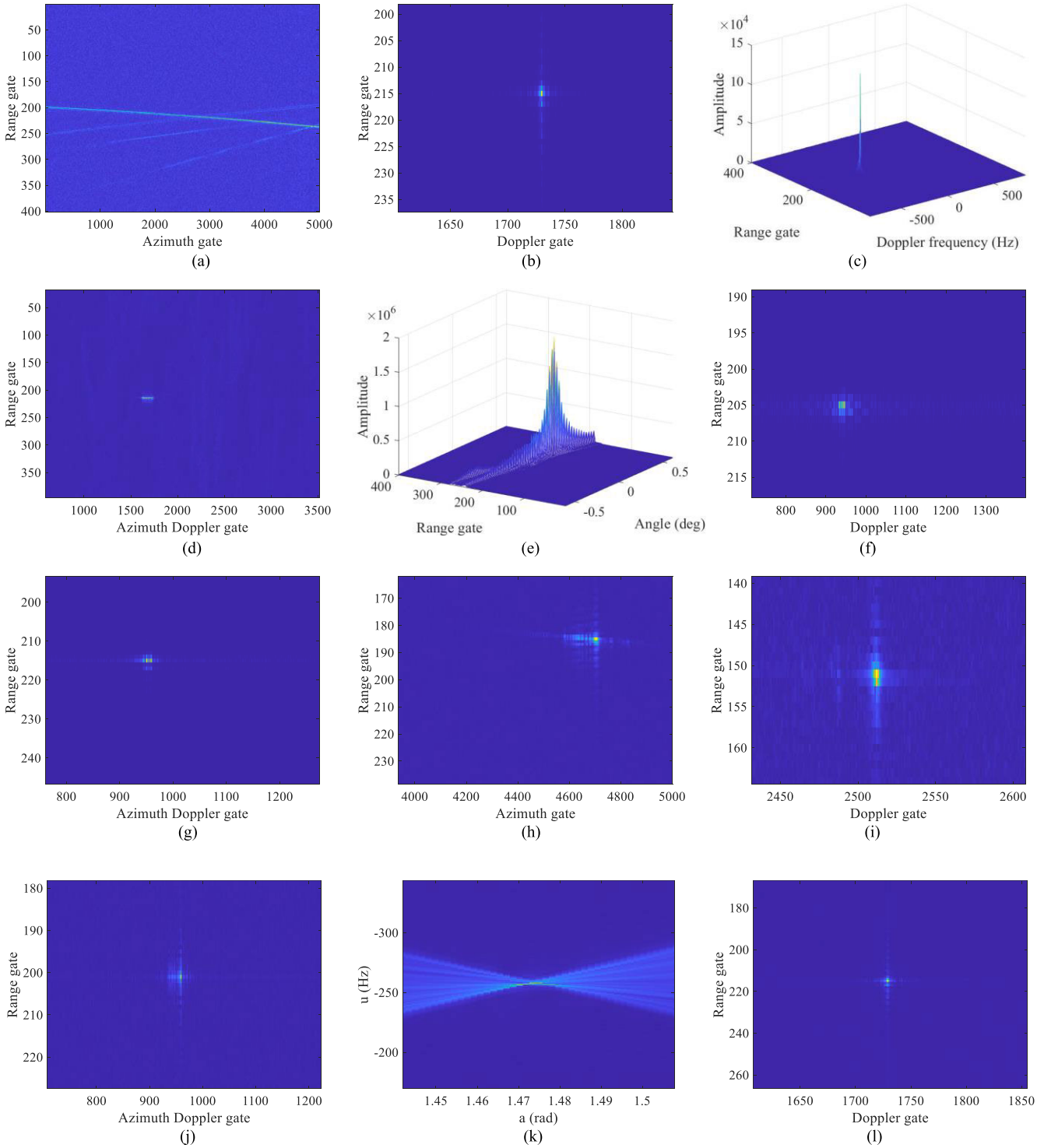


Fig. 11. Moving target focusing on a four-channel SAR system. (a) Target motion trajectory. (b) Target imaging result by using the proposed approach. (c) Stereogram of target focusing result with respect to (b). (d) Target imaging result by using the DKP method. (e) Target rotation angle searching result by the IAR-TRT. Target imaging result by using (f) IAR-TRT and (g) KT-TRT method. Imaging result by (h) 2-D frequency MF approach, (i) BFCA approach, (j) FRFT approach, (k) Radon-FrFT approach, and (l) KTMF approach.

the asymmetrical spectrum broadening appears, as displayed in Fig. 9(f)–(j). As an optimal parameter estimation method with a 2-D grid search, this target is finely imaged by the KTMF approach, as presented in Fig. 9(k) and (l). From the figures, we can see that the focusing performance of this target by using the proposed approach is similar to that of the KTMF approach, but with much low computational complexity. In summary, the proposed approach can realize a good balance between target imaging quality and computational resource consuming. Unfortunately, due to the fact that only two receiving channel data are available for this SAR system to implement the adaptive clutter filtering, the additional spatial-domain channel resource is deficient for the subsequent target relocation.

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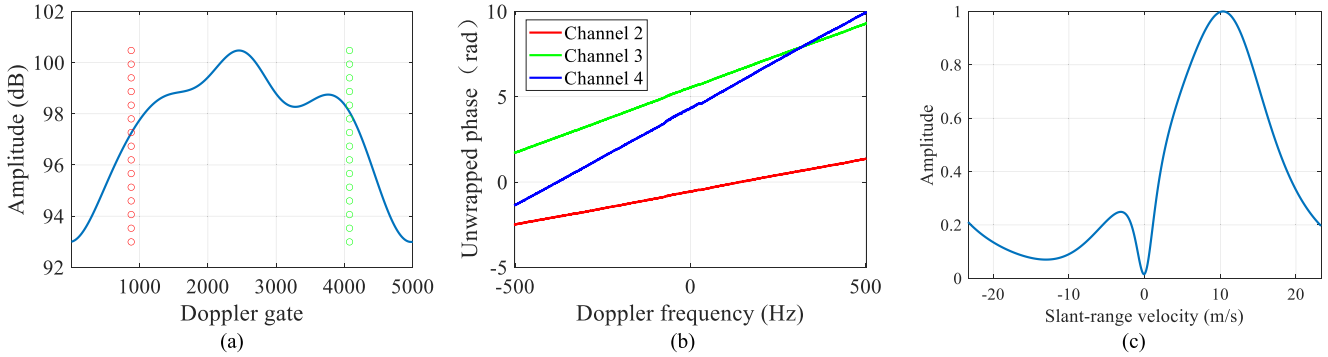


Fig. 12. Along-track baseline estimation and target radial velocity estimation results. (a) Interferometric phase distributions between channel 1 and rest channels. (b) Interferometric phase distributions. (c) Target slant-range velocity estimation by the DRAMF method.

TABLE VII
C-BAND SAR SYSTEM PARAMETERS

Range bandwidth	210 MHz
Range sampling frequency	267 MHz
Pulse repetition frequency	1673 Hz
Pulse duration time	10.4 μs
Platform velocity	131 m/s
Platform height	8600 m
Nearest slant range	10.5 km
Squint angle (degree)	0

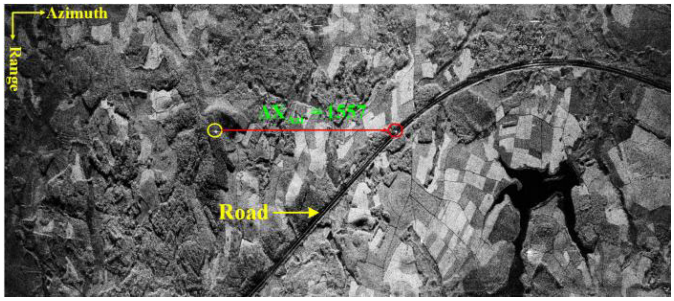


Fig. 13. Ground target relocation result in a four-channel high-resolution SAR system.

Later, another newly recorded airborne SAR data is displayed to verify the proposed approach, which was recorded by a high-resolution C-band radar with four azimuth receiving channels and the observed scene belongs to the mountainous area of China. The observed target is a ground moving car, the nominal distance between two adjacent channels is 0.15 m, and the other detailed parameters of the SAR system are provided in Table VII. The well-imaged result of ground stationary scene is depicted in Fig. 10, where the defocused target signal tagged by a yellow ellipse will be applied for the subsequent focusing performance evaluation.

After clutter rejection, Fig. 11(a) shows a curved motion trajectory of the target signal. Due to the effective high-order parameter estimation and smeared phase term compensation, this ground target is finely integrated by the proposed approach and the KTMF, as presented in Fig. 11(b), (c), and (l). Conversely, since the phase error induced by c_3 is not well dealt with in this high-resolution SAR data, Fig. 11(d)–(k) displays the target asymmetrically broadening results by using the DKT method, IAR-TRT method, KT-TRT method, 2-D frequency MF method, beat frequency coherent accumulation (BFCA) method [3], FRFT method, and Radon-FrFT method. Thus, the proposed approach can be applicable to effectively and efficiently accomplish the ground target imaging in this high-resolution SAR system.

After target refocusing according to the estimated motion parameters, in the following, the target radial velocity is estimated by utilizing the adaptive array signal processing based on four spatial receiving channels. For this airborne SAR system, although the nominal baseline is given, the practical baseline will deviate from the ideal baseline to some extent due to the behaviors of the wind, which is adverse to the target cone angle or radial velocity estimation. Hence, the along-track baseline should be re-estimated according to the receiving radar echo data. Fig. 12(a) shows the statistical clutter amplitude along range dimension, and the 3-dB mainlobe clutter components labeled by red and green circles will be exploited to accomplish the baseline estimation. The unwrapped interferometric phase distributions among the 3-dB mainlobe regions between the reference channel (channel 1) and the rest channels are depicted in Fig. 12(b), from which one can see that three beelines with the inclined degree related to the along-track baseline are generated. According to the slope rate in Fig. 12(b), the along-track baseline can be well estimated by exploiting the linear least-square fitting technique, with the baseline estimation results given by $[\hat{d}_1, \hat{d}_2, \hat{d}_3, \hat{d}_4] = [0, 0.167, 0.3148, 0.4781\ \text{m}]$. Finally, based on the estimated baseline, the target slant-range velocity estimation result by using the adaptive data reconstructed AMF (DRAMF) [36] is given in Fig. 12(c), where v_r can be estimated as 10.3476 m/s.

Based on the radial velocity estimation result in Fig. 12(c), the moving target discrete displacement along azimuth dimension can be estimated by $\Delta X_{\text{azi}} = \text{round}(2T_a\hat{v}_r/\lambda)$, i.e., $\Delta X_{\text{azi}} = 1557$. Finally, Fig. 13 depicts the target reposition result, from which it can be clearly observed that the target can be accurately repositioned on the road.

C. Multichannel Spaceborne Radar Simulation and Analysis

In this section, the real-measured spaceborne SAR data is exploited to evaluate the processing performance of the

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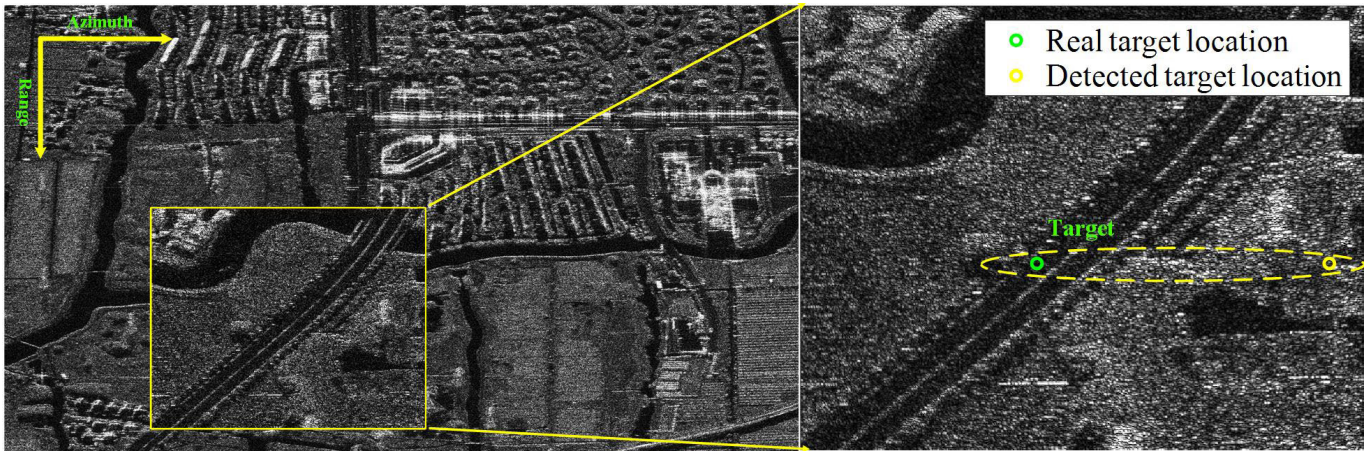


Fig. 14. Gaofen-3 spaceborne SAR imaging result and moving target position setting in a four-channel SAR-GMTI experiment (the experiment region covers the Ying Gang East Road of Shanghai, China).

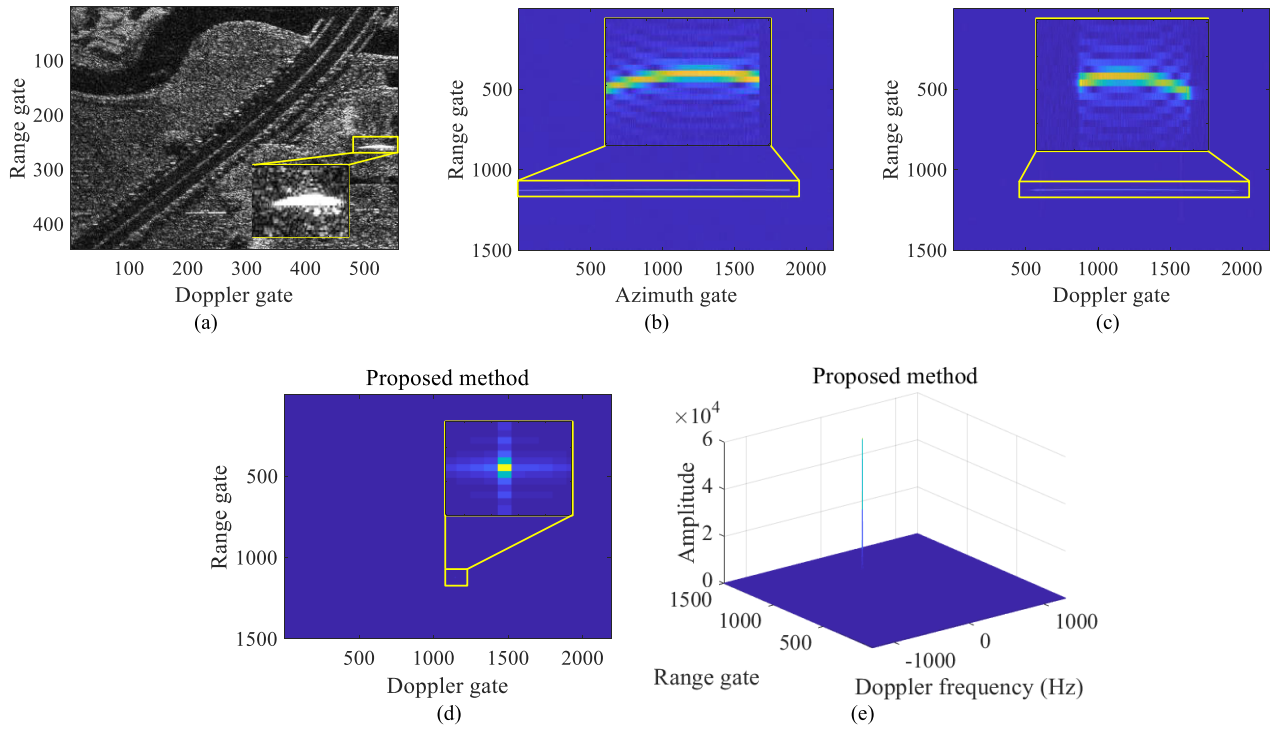


Fig. 15. Moving target focusing on a spaceborne multichannel SAR system. (a) Smeared target imaging result in an SAR image. (b) Target trajectory in the slow-time domain. (c) Target frequency spectrum in the Doppler domain. (d) Target imaging result by using the proposed approach. (e) Stereogram of target focusing result.

proposed approach, which was recorded by the Chinese Gaofen-3 SAR satellite. Considering that only two receiving channels are installed in this spaceborne SAR system, no more spatial degree of freedom is available to realize the ground moving relocation and clutter rejection, simultaneously. To deal with this issue, two more spatial channels are simulated according to the clutter spatial vector distribution, where the adjacent channel distance is 7.5 m and the other SAR parameters are given in Table VIII. The imaging scene is displayed in Fig. 14, where the road region named Ying Gang East Road nearby Shanghai Hongqiao airport of Shanghai, China is chosen as the clutter scene (see the yellow rectangle box) to test the ground target focusing and relocation performances. In this semireal measured SAR data, a ground maneuvering target with $v_x = -20.11$ m/s, $v_y = 25$ m/s, $a_x = -5$ m/s^2, and $a_y = -3$ m/s^2 is added. The target true position and the shifted position are labeled by the green circle and yellow circle, respectively. Fig. 15 shows the processing results for this target.

Fig. 15(a)–(c) shows the smeared target imaging result, the target trajectory in the slow-time domain after clutter rejection, and its frequency spectrum, respectively. It can be seen from the figures that, due to the uncooperative motions, this target is typically smeared and shifted. After performing the target Doppler parameter estimation and motion compensation (see the parameter estimation results in Table IX), this moving

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Fig. 16. Localization results in a Google Earth image, where the real target location is labeled by a red marker and the relocated target position is labeled by a blue marker.

TABLE VIII
SYSTEM PARAMETERS

System bandwidth	50 MHz
Pulse duration	25 us
Pulse repetition frequency	2588 Hz
Wavelength	0.056 m
Platform velocity	7569.5 m/s
Platform height	755 km

TABLE IX
TARGET PARAMETER ESTIMATION RESULTS

Parameters	Second-order coefficient	Third-order coefficient	Radial velocity
Real value	31.9903 m/s^2	0.0133 m/s^3	-9.5736m/s
Estimated value	31.9901 m/s^2	0.01018 m/s^3	-9.3936m/s

target is finely imaged by the proposed method, as shown in Fig. 15(d) and (e). Finally, after implementing the target radial velocity estimation by the DRAMF method, this simulated target is precisely relocated on the road in the Google Earth image by using the keyhole markup language files, as provided in Fig. 16. Therefore, the proposed method can also be employed to realize the ground target focusing and relocation in the semireal measured spaceborne SAR scene.

V. CONCLUSION

To deal with the ground maneuvering target smearing issue in a high-resolution SAR system, an effective and efficient parameter estimation and target imaging approach according to the symmetric characteristics of slow-time and range frequency have been studied in this article. The main contributions include the following: 1) the RCM including range walk and range curvature and the complex Doppler phase errors induced by the unknown motions of a ground maneuvering target can be well compensated; 2) the proposed approach is suitable for the multitarget case where the complex coupling effects still exist in the cross-terms after accomplishing the phase decoupling with respect to the autoterms; and 3) the proposed method requires much less computational resource since the time-consuming grid search procedures are avoided. The effectiveness of the proposed approach has been verified by numerous simulation and real-measured experiment examples.

Because of the nonlinear transform property of MSF transform, the proposed approach is more suitable for the efficient refocusing of ground maneuvering targets with relatively strong intensities. In addition, when only two azimuth receiving channels in an SAR system are available, the moving target relocation procedure may not be effectively accomplished after clutter rejection. How to realize the precise signal separation for ground continuous clutter and the robust clutter suppression via the advanced radar waveform and algorithm designment as well as efficient ground target imaging under the low SCNR situation in a new-style airborne or spaceborne multi-input and multiple output radar system will be under our future investigation.

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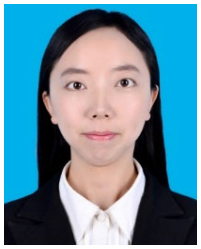
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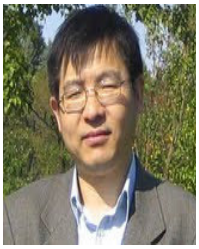


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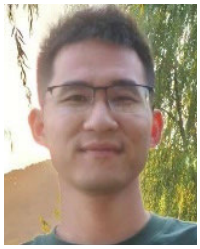
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